SmartGrid AI: End-to-End Architecture and Risk Communication

Reference workflow for ML predictions, RAG explanations, and dashboard presentation

Document purpose: This document is prepared as a searchable knowledge-base source for a SmartGrid AI RAG prototype. It is an educational synthesis, not an official ERCOT, utility, or government publication.

System objective

SmartGrid AI is a decision-support concept that combines electricity-demand forecasting, grid-stress classification, document retrieval, language generation, and a dashboard. Its goal is to help users understand possible periods of elevated system pressure and the context behind a model estimate.

The system is not a grid control system and should not issue operating instructions to utilities. Official operator and utility communications remain authoritative.

Recommended workflow

1. Data preparation: clean and align historical demand, weather, outage, and geographic data. Document data coverage and units.

2. Regression: predict the selected future demand target, such as next-day peak demand or hourly load.

3. Classification: estimate the defined grid-stress class using the appropriate features and target labels.

4. RAG retrieval: search trusted documents using the model output and relevant context.

5. LLM explanation: generate a concise explanation grounded in retrieved passages and clearly marked model estimates.

6. Dashboard: display forecast values, class, score, forecast horizon, evidence sources, limitations, and practical guidance.

Dashboard design

Show the prediction timestamp and forecast period prominently. Display demand with its unit and distinguish predicted values from observed historical values. Show the classification label and score with a clear explanation of what the score represents.

Include an explanation panel with source links or document references. Include a consumer-advice panel that avoids alarmist language. Provide a visible note that the prediction is not an official grid alert and that users should follow their utility and ERCOT communications.

Use charts to show historical demand, forecast demand, and relevant weather variables. Avoid charts that imply causal relationships unless the analysis supports them.

Risk communication principles

Use terms such as 'model-estimated elevated risk' rather than 'blackout certain'. Explain uncertainty and the forecast horizon. Separate observed facts, model predictions, retrieved document evidence, and generated interpretation.

Do not interpret a classifier score as the chance of an outage unless the target label is an outage and the model has been properly calibrated and validated for that interpretation.

Recommendations should be proportionate and safe. Encourage flexible conservation only when appropriate, and never tell users to compromise medically necessary equipment or essential heating and cooling.

Operational safeguards

Record model version, data timestamp, feature schema, and RAG document versions. Validate input fields and handle missing values. If the ML service is unavailable, show that the prediction is unavailable rather than generating a fabricated result.

If retrieval returns no relevant evidence, the LLM should clearly say it cannot find sufficient supporting information. If the LLM service fails, the dashboard should still display the ML output with a clear message that the explanation is unavailable.

Periodically review documents for outdated guidance, broken source links, and conflicting information. Keep official alerts separate from generated summaries.

Key takeaways

- Keep regression, classification, retrieval, generation, and dashboard responsibilities distinct.
- Display model estimates separately from official alerts and documented facts.
- Show forecast horizon, units, timestamp, sources, and limitations.
- Design graceful failure behavior for missing predictions or weak retrieval.

References and further reading

• ERCOT Grid Information — https://www.ercot.com/gridinfo

• ERCOT Demand Response — https://www.ercot.com/services/programs/load

• NERC Reliability Assessments — https://www.nerc.com/our-work/assessments

Prepared 9 November 2026. Verify current operational information with the linked official sources before making real-world decisions.